



Republic of the Philippines  
Department of Agriculture  
BUREAU OF FISHERIES AND AQUATIC RESOURCES  
**REGIONAL FISHERIES LABORATORY XII**  
J. Catolico St., Lagao, General Santos City

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|------------------------------------------------------|-----------------------------------------|------------------------------|-------------------------|
| Document Type<br><b>Standard Operating Procedure</b> | Revision No:<br>0                       | Date Adopted:<br>13 Aug 2019 | Page No:<br>Page 1 of 5 |
| Document Code:<br><b>SOP-09</b>                      | <b>ENSURING THE VALIDITY OF RESULTS</b> |                              |                         |

## Introduction

This procedure describes the measures undertaken for ensuring the validity of test results according to the requirements of the sub clause 7.7 of the PNS ISO/IEC 17025:2017.

## 1. Responsibilities

1. Laboratory Manager
2. Quality Assurance Manager
3. Technical Manager
4. Laboratory Analysts

## 2. Quality Control

2.1. Routine controls to assure the quality of the test results include:

1. Retesting of test items before the release of results to the customers by the same and/or other staff of the laboratory.
2. Replicate tests
3. Use secondary reference materials, if necessary.
4. Participation in proficiency testing schemes and/or inter-laboratory comparison.

2.2. Quality controls are carried out for every analysis/tests done by the Laboratory Analyst. The results are recorded according to the respective test procedures in the form LF 09-01, Quality Control Table.

2.3. In case of a retest, the results are assessed by the Technical Manager and discussed with the Laboratory Manager and Quality Assurance Manager. If deviations are outside the acceptable limits, a cause analysis may be carried out to find out possible causes. One of the actions to find out possible causes is to repeat the analysis, including the quality control. The findings of any quality control are recorded in the LF 10-01, Non Conformity Form.

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| Document Code:<br><b>SOP-09</b>                      | <b>ENSURING THE VALIDITY OF RESULTS</b> |                              |                         |

### 3. Quality Control Acceptance Criteria

#### 3.1 Internal Quality Control

Included are the following Quality Control techniques in each analytical method or WI:

- Reagent Blank- to determine the contributions of the reagents and preparative analytical steps to error in the measurement.
- Replicate Samples - to determine the repeatability or precision of the analytical method.
- Spike samples – to calculate % recovery.
- Certified Reference Materials if available
- Calibration standards – to prepare standard curves and to verify if the instrument calibration has been maintained.
- After the run, calculate the % Recovery rate for the parameter of interest.

#### Quality Control for PCR

##### 3.1.1 Reagent Blank (Histamine)/Negative Blank (PCR)

3.1.1.1 Refer to WI-01-CHE-02, Histamine, for preparation of reagent blank.

3.1.1.2 Refer to WI-01-FIS-01, Molecular Diagnosis, for preparation of negative blank.

##### 3.1.2 Replicate Samples

3.1.2.1 Refer to WI-01-CHE-02, Histamine, and WI-01-CHE-03, Moisture Content, for chemical analysis.

3.1.2.2 Refer to WI-01-MIC-02 to WI-01-MIC-08 for microbiological analysis.

3.1.2.3 Refer to WI-01-FIS-01 for Molecular Diagnosis.

##### 3.1.3 Spike Samples

3.1.3.1 Refer to WI-01-CHE-02, Histamine, for preparation of spike samples.

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| Document Code:<br><b>SOP-09</b>                      | <b>ENSURING THE VALIDITY OF RESULTS</b> |                              |                         |

### 3.1.4 Calibration Standards

3.1.4.1 Refer to WI-01-CHE-02, Histamine, for preparation of calibration standards.

### 3.2 Quality Control Acceptance criteria

This includes:

3.2.1 Quality Control Chart- is a tool use to monitor a process. The data used in this method are the data gathered from results of analyses (see guidelines in SOP-09-01, Quality Control Chart Guide) depending on which quality control chart the analyst is to prepare. These data are derived from the data sets used to statistically derive the standard deviation of the method.

- Warning Limit- Usually set at  $\pm 2$  std dev.
- Control Limit - Usually set at  $\pm 3$  std dev.

#### 3.2.2 Evaluating Quality Control Chart

A. When a non conformity is encountered, the analyst must perform the following;

- Repeat the analysis;
- Check routine instrument maintenance procedures;
- Recheck quality of materials, reagents;
- Review procedures.

B. When a non-conformity never or rarely happens, recalculate the warning and control limit using 20 most recent data points.

### 3.3 External Quality Control

3.3.1 Arrange for the laboratory to participate in Interlaboratory Comparison (ILC) and/or proficiency testing schemes (PTs).

3.3.2 Evaluate the results of ILC or PTs.

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| Document Type<br><b>Standard Operating Procedure</b> | Revision No:<br>0                       | Date Adopted:<br>13 Aug 2019 | Page No:<br>Page 4 of 5 |
| Document Code:<br><b>SOP-09</b>                      | <b>ENSURING THE VALIDITY OF RESULTS</b> |                              |                         |

### 3.4 Corrective Action

- 3.4.1 Fill up form – LF 10-01, Non Conformity Form
- 3.4.2 Determine actions needed to correct the problem and institute remedial/ corrective action.
- 3.4.3 Implement the corrective action.
- 3.4.4 Verify the effectiveness of the corrective action.

## 4. Participation in Proficiency Testing schemes (PTs) or Inter-Laboratory Comparisons (ILC)

- 4.1. Participation in proficiency testing schemes or inter-laboratory comparisons takes place every two years. Successful participation of any laboratory in an inter-comparison scheme is decided upon approval by the Laboratory Manager.
- 4.2. Comparisons of inter-laboratory results are undertaken to determine the technical capabilities of the laboratories.
- 4.3. At the end of each inter-comparison scheme and after publication of the results, the respective head of each laboratory submits a report to the Laboratory Manager and Quality Assurance Manager by including information with respect to the performance of the laboratory.
- 4.4. Participation in an inter-comparison scheme is recorded in the LF 09-02, Participation in PTs and/or ILC.
- 4.5. The participation and the performance in PTS and/or ILC measurements are included in the management review.
- 4.6. Proficiency test plan (LF 09-04) will be filled out by respective technical managers of each unit.

## 5. Documents

- LF 09-01, Quality Control Table
- LF 09-02, Participation in PTS and/or ILC
- LF 10-01, Non Conformity Form

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WI-01-CHE-02, Histamine  
WI-01-FIS-01, Molecular Diagnosis in Shrimp  
WI-01-MIC-02 Heterotrophic Plate Count  
WI-01-MIC-03 Aerobic Plate Count  
WI-01-MIC-04 Coliforms and E. coli in Water and Ice  
WI-01-MIC-05 Staphylococcus aureus in Fish and Fishery Products  
WI-01-MIC-06 Coliforms and E. coli in Fish and Fishery Products  
WI-01-MIC-07 Salmonella  
WI-01-MIC-08 Shigella

## 6. Document History

| Revision No. | Issue Date | Original Entry | Revised as | Recall Date |
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